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Negentropic effects in quantum-based noise devices during biofield healing

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ABSTRACT

One thousand custom-designed quantum noise generators (QNGs) were employed in a biofield healing experiment as exploratory sensors for a purported biofield. The devices continuously recorded random voltages in electronic components at 10 Hz, where the randomness was based on quantum tunneling and other physical sources of noise in semiconductors. Each of six healing sessions was about 30 min long, preceded and followed by short baseline periods with and without the healer present in the laboratory. Analysis of average noise levels, compared with control data, indicated significant deviations from chance expectation during the healing periods. A secondary analysis quantifying fractal complexity confirmed a significant negentropic shift specifically during the healing periods and compared to matched control sessions. This outcome is consistent with results observed in previous healing studies, suggesting that energy medicine healing is associated with a negentropic effect in the environment that can be detected as the emergence of unexpected order in quantum-based randomness.

Introduction

Biofield healing modalities, known variously as energy medicine, subtle energy healing, Reiki, qigong, Therapeutic Touch, etc., assume that a practitioner's intentions and non-contact interactions induce a healing response in a client. These techniques reflect diverse cultural origins, distinct terminologies introduced by their developers, and differing assumptions about underlying mechanisms, including notions of universal or vital energies. Despite different terminology and cultural framing, these approaches share the central assumption that directed intention or consciousness can influence biological systems through mechanisms not yet understood. Whether such effects are mediated by conventional physiological pathways, electromagnetic fields, or quantum processes, or as-yet-uncharacterized biofield interactions, remains an open empirical question warranting systematic investigation.

The goal of this study was to develop a sensor system capable of detecting and visualizing purported informational fields associated with biofield healing.[1–3] If successful, such a system could offer a possible means for enhancing biofield healing by providing practitioners with feedback and an objective measure of the biofield itself. Previous approaches have included algorithmic analysis of photographic images, corona discharge imaging of the skin, measurements of molecular or pH changes in water, variations in electromagnetic or magnetic fields, and detection of biophotons.[4,5] Such efforts have produced intriguing results, but their reliability and clinical applicability remain largely speculative and debated.[6–10]

One reason for uncertainty in detecting a purported biofield with conventional instruments is that the term "energy," as often used in this context, may not correspond to the physical definition of energy in physics. Instead, healing interactions may be associated with changes in information, or patterns carried by energetic processes. The difference between energy and information in this context can be illustrated by whispering to a friend either of the following phrases: "I love you" or "you smell bad." Both cases may involve identical amounts of sonic energy, but the difference in informational content can evoke dramatically different reactions.

To objectively measure effects of a purported informational "organizing" field, an appropriate instrument must generate maximal entropy and detect deviations from it.[11,12] One of the first such devices was created for measuring subtle mind-matter interaction effects, such as those purportedly involved in biofield healing.[13] These devices are known as truly random number generators (RNG), where the adjective *truly* indicates that the source of randomness is a physical process rather than a software algorithm. The random bit samples produced by most commercially available RNGs are conditioned ("whitened") to yield statistically unpredictable sequences of 0 s and 1 s. These sequences are verified to ensure that the bits are independent and identically distributed (i.i.d.), using established analytical test suites.

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Previous studies suggest that when individuals or groups engage in highly focused attentional states, such as during compelling musical or athletic events, negentropic effects in RNGs can occur.[14–18] In addition, when one or more RNGs are placed in the vicinity of biofield healing sessions, similar negentropic outcomes have been measured.[15,19–22] Here we describe the use and results of a novel type of RNG used as part of an exploratory biofield visualization project.

Method

Three biofield healers participated in six healing sessions, each conducted with a different volunteer reporting anxiety.[23] All volunteers provided written informed consent under approval by the Institutional Review Board of the Netherlands Organization for Applied Scientific Research (TNO, Soesterberg, Netherlands). Volunteers were recruited from the TNO participant pool, and healers were identified through professional biofield therapy networks. Healers were selected based on pre-specified inclusion and exclusion criteria and an interview that included a demonstration session with a study team member. Participants received remuneration of €65 plus travel expenses and healers received compensation of €150 euros plus travel expenses.

Each session involved one biofield therapy practitioner (BTP) and one healee in a one-to-one format. Each BTP conducted two sessions with different healees. The BTP was seated approximately 4 m from the healee, who was continuously recorded on video. Two 5-minute baseline periods were collected before healing, one without and one with the healer present, followed by a non-contact healing session conducted according to the practitioner's usual method, lasting up to 25 min. To minimize electromagnetic interference, BTPs were instructed to remain seated, avoid physical contact, and restrict movement to gentle hand motions if customary to their practice. Verbal communication was limited to brief, essential exchanges. Throughout the session, the principal investigator recorded timestamps and observations, and a secondary researcher monitored external conditions. The BTP determined the session length, with a reminder issued at 20 min to conclude by 25 min.

Both BTPs and healees completed a battery of psychological and phenomenological assessments immediately before and after the healing, including the Self-Assessment Manikin (SAM), Inclusion of Self and Others (IOS), Perceived Body Boundaries Scale (PBBS), Spatial Frame of Reference Continuum (S-FoRC), and the Nondual Awareness Dimensional Assessment (NADA). Healees additionally provided qualitative descriptions of their experiences, while BTPs described their healing process in short audio-recorded interviews. These results will be reported elsewhere. Pre- and post-session procedures (check-in, setup, questionnaires, and debriefing) followed a standardized protocol lasting approximately 2 h. A follow-up interview and anxiety assessment (STAI) were conducted one week later.

During the healing sessions, multiple physical and physiological sensors operated continuously and simultaneously, including autonomic and electrocortical measures, biophoton and infrared cameras, and the focus of this paper, electronic circuits designed to generate truly random samples. To synchronize the data from these diverse devices, the Lab Stream Layer software framework was employed.[24] Separate manuscripts describing the results from the remaining sensors are being prepared.

Equipment

The custom-designed sensor system consisted of 100 quantum noise generators (QNGs) arranged in a 10 × 10 matrix on a printed circuit board approximately one foot square (fabricated by ubld.it, Chase, MD). The "Q" in QNG refers to a quantum-indeterminate source of randomness, in this case electron tunnelling in a reversed-biased Zener diode. These semiconductor components also exhibit three other sources of noise: (1) Johnson–Nyquist noise from thermal effects in electronic components, (2) shot noise from current fluctuations due to the discrete charge of the electron, and (3) avalanche noise arising from the unpredictable motion of many electrons. The combination of these four noise sources was used to create a truly random sequence of samples, and instead of converting samples into binary bits, as in most commercial RNGs, the voltage returned by the Zener diode was digitized to 12 bits of resolution and these digitized samples were generated at 10 Hz. Each QNG board thus generated 1000 random samples per second. These samples were sent via USB to a computer. Fig. 1 shows one of these boards.

The reason we used QNGs in this study, rather than RNGs, was to explore why the bit samples generated by RNGs in previously studies deviated from chance. Was it something about the quantum noise itself that was altered, or something about the "whitening" process in which noise was converted into random bits, or some other effect that influenced the bits after they were already created? If this experiment suggested that quantum-level noise itself was modulated, then that might provide a hint about the underlying mechanisms of healing intention, namely that it may have had something to do with quantum phenomena, such as the quantum observer effect.[25]

Ten such QNG boards were employed in this experiment. As shown in Fig. 2, six boards were placed behind the chair where the volunteer sat, two were positioned on either side of a table in front, and the remaining two (not shown) were located about three meters away at the opposite end of the room. These 10 boards together generated 10,000 random noise samples per second.

Analysis

In the following, the term "sample" refers to an individual voltage measurement of random noise taken every tenth of a second from one of the 100 QNGs on each board. Fig. 3 shows 50,000 successive measurements. Each value represents the mean of 10,000 samples produced by 1000 QNGs across the ten boards every tenth of a second, along with a histogram and an autocorrelation to lag ±100.

These data were recorded during a baseline sampling period with people in the lab and all of the various sensors running, but with no healing activities taking place. Analysis indicated that the QNG samples conformed to the statistical behavior expected of truly random data. This was further confirmed by subjecting the raw data to a Wald-Wolfowitz runs test (Matlab function *runstest*) and to a Ljung-Box autocorrelation test[26], both of which failed to detect any non-random sequential behavior.

To see if this random noise was affected by biofield healing, samples were continuously collected during each of six half-hour sessions. Each session comprised five stages: (1) three minutes with the volunteer alone in the laboratory, (2) three minutes after the healer entered, (3) a 20-minute healing session, (4) three minutes post-healing, and (5) three minutes after the healer's departure.

Each session was comprised of approximately 20,000 random samples recorded at 10 Hz (≈2000 s or 33 min). We may note that the internal sample rate of each QNG was 10KHz, but we were interested in effects that would likely manifest at a "human scale," i.e., over minutes, as observed in previous healing studies, rather than in high-frequency bursts. For that reason, and for the sake of computational efficiency, we opted for a slower sample rate.

Timing varied slightly across sessions; therefore, to ensure time synchronization, the data were interpolated so that each event contained the same number of samples (MATLAB function *interp1*). To offset potential autocorrelations created by the interpolation, the data array was downsampled by a factor of 2 to provide the equivalent of a low-pass filter, and thus the resulting samples were recorded at the equivalent of 5 Hz. Subsequent Wald–Wolfowitz runs tests and Ljung–Box autocorrelation tests applied to each 10,000-sample data array showed no evidence of sequential dependence.[26,27]

Next, the 100 data arrays from each board were concatenated into a

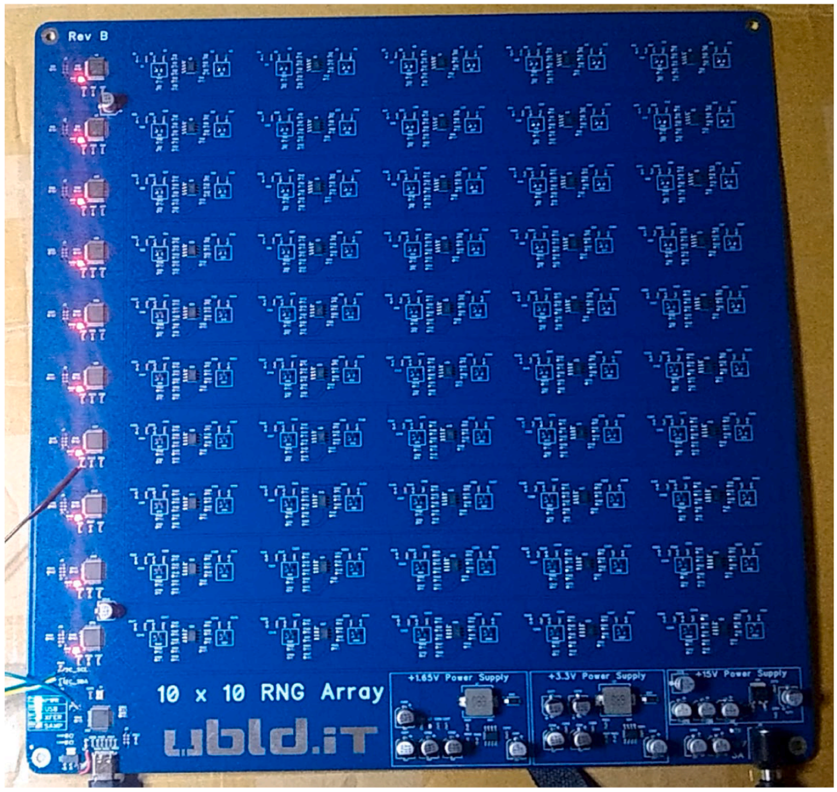

Fig. 1. Photo of a QNG board. See text for detailed description.

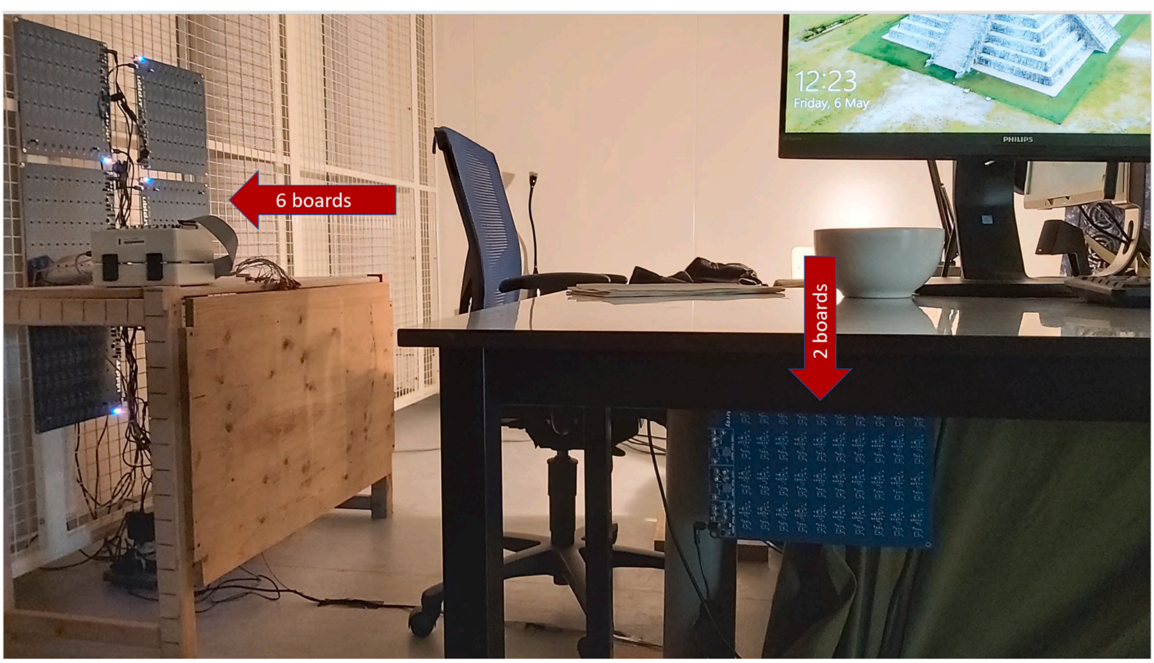

Fig. 2. Location of 8 of the 10 QNG boards in the laboratory test.

100 × 10,000 matrix. Then the mean across the 100 noise values per sample was determined and smoothed by a 5-minute moving mean to provide a low-pass filter to help visualize slower moving deviations in the QNG outputs. This was repeated for each of the 10 boards per session. Each of the resulting 60 data arrays (10 boards x 6 sessions) was individually tested for sequential randomness using the Wald-Wolfowitz and Ljung-Box, for a total of 120 tests.

To assess how each smoothed data array, and the combined data across all boards and sessions, deviated from chance expectation, the following nonparametric approach was applied: The data array from each session was randomly permuted, then, as before, a sample mean was formed across the 10 boards for each of the 10,000 samples, and then the resulting array was smoothed with the 5-minute moving mean. This randomization procedure was repeated 5000 times, storing the

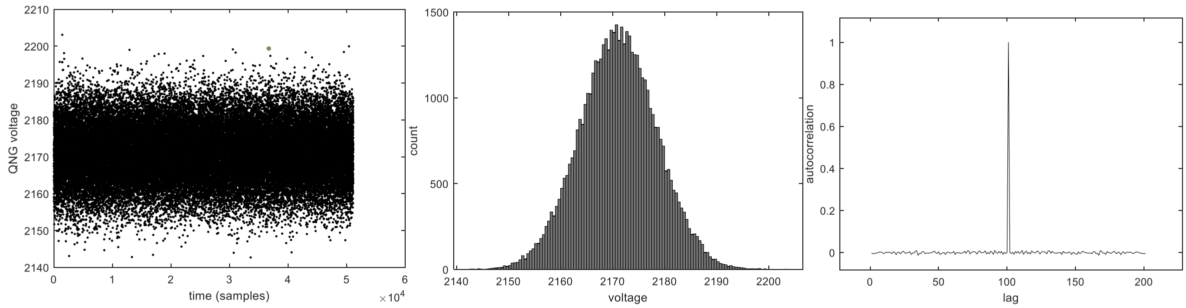

Fig. 3. (Left) Means of 10,000 samples generated every tenth of a second across the 10 QNG boards, measured continuously over 83 min. (Middle) Histogram of these means. (Right). Autocorrelation analysis lag ±100, showing no temporal dependency between the mean samples.

randomized array each time.

Each sample in the original smoothed data array was then compared to the 5000 randomly shifted values for that sample, and then $p_i = \sum (k_i \geq m_i)/N$, where p was the probability that the i th sample deviated from the corresponding sample mean across the randomly shifted data arrays, $N = 5000$, and i ranged from 1 to 10,000. To avoid uninterpretable results when performing an inverse normal transform on the resulting p valued, if $p_i = 0$, then $p_i = 1/N$, and if $p_i = 1$, then $p_i = (N-1)/N$. Then $z_i = norminv(p_i)$. This procedure was repeated for all 10,000 samples per data array, and where z was now a standard normal deviate. This process was repeated for all data across the six healing sessions, and then combined into a single curve as a Stouffer Z score (which is also distributed as a standard normal deviate),[28] $SZ = \sum z_{ij}/\sqrt{60}$, where i refers to the 10 boards per session, j to the 6 sessions, and 60 to the total number of curves that were combined.

To provide a control comparison, data generated by the same QNG boards during cognitive and perceptual tests, with no healing taking place, were used. Of 24 such sessions, six sessions were selected where their baseline QNG measures matched the baselines observed in the six healing sessions as closely as possible. These sessions were longer than the healing sessions, so the first 20,000 samples were used to provide "pseudo-healing" controls. All analyses applied to the healing session data were applied identically to the data from the six control sessions.

Results

Fig. 4 shows the results. Using the False Discovery Rate (FDR) procedures to adjust for multiple tests with α = 0.05 indicates that for about 5 min at the start of the healing session the mean noise level declined nearly 4 σ below chance expectation, then it eventually increased during the healing period to peak at nearly 4 σ above chance a few minutes before the healing session ended.[29] The control data also showed a decline in mean noise near the start of the healing session, but it did not survive FDR correction. A visual correlation between the healing and control curves suggested the presence of common environmental influences, but the curve for the controls did not survive adjustment for false discovery rate whereas the latter did. This suggests that the healing session results overcame any common environmental effects.

To test this idea, we performed two complementary analyses on the phase-aligned curves (20,060 samples each, pooled across N = 6 healing and N = 6 matched-control sessions). First, we fit an ordinary least-squares regression of the healing Stouffer Z curve on the matched-control Stouffer Z curve across all aligned samples, obtaining SZ_heal = 0.023 + 1.170 × SZ_ctrl with a Pearson correlation of $r = 0.809$. This strong correlation confirmed that a substantial fraction of the shape of the healing curve was shared with the control curve, consistent with common environmental influences, while the regression slope greater than 1 implied that the healing curve's excursions were modestly amplified relative to controls. Subtracting the fitted line yielded a residual curve representing the component of the healing signal that was orthogonal to the matched-control signal.

Second, to see if the strongest features of the original healing curve remained statistically reliable after removing the control contribution,

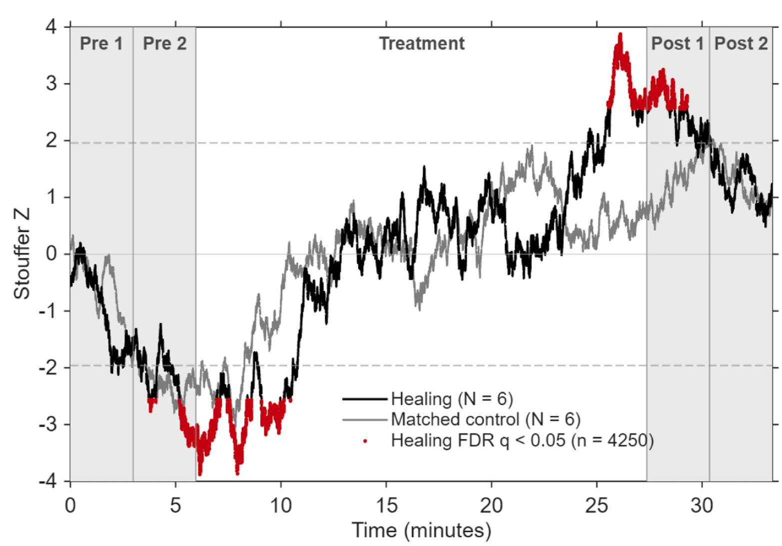

Fig. 4. Results of original data combined over 10 QNG boards × 6 sessions compared to 5000 randomly circular shifted permutations of that data, for healing sessions (black line) and matched control sessions (grey line). The red portion of the healing curve indicate statistically significant results at p < 0.05 after adjustment for false discovery rate; no portion of the control curve survived that adjustment.

we paired each healing session with its closest-baseline matched control, computed per-sample differences, and then combined them across the six pairs via a Stouffer Z. We then restricted FDR correction to the 100 samples with the largest $|Z|$ in the original healing Stouffer curve ($|Z| \geq 3.68$), the regions where the healing effect was most pronounced (rather than correcting over the full ~4250-sample healing window, which would have imposed an unnecessarily severe multiple-comparisons penalty on locations that were never candidates for the effect). Of those 100 peak samples, 45 (45.0%) survived FDR in the paired-difference test, indicating that nearly half of the strongest healing excursions remained significantly different from their matched controls.

To further investigate how the act of healing may have altered the temporal structure of quantum noise, for purely exploratory reasons we analyzed the Higuchi fractal dimension (HFD) associated with the QNG time series data for both the healing and control sessions.[27] This metric quantifies the complexity of a time series by measuring the fractal dimension at multiple time resolutions. A perfectly smooth, predictable signal, like a sine wave, would have a low fractal dimension, close to 1.0. By comparison, purely random white noise would result in HFD ≈ 2.0. HFD essentially asks: How much does this signal twist and turn on itself? Or, how self-similar is its structure across different timescales? HFD is useful in the present context because it is particularly sensitive to subtle changes in complexity that cruder measures (like mean noise levels) would not detect.

Results of this analysis showed that 5 of the 6 healing sessions showed lower HFD during the healing period compared to the same period in the matched controls (mean $\Delta = -0.0021$, SD = 0.0022). Nine of the 10 QNG boards showed negative HFD differences during the healing period. There was no significant correlation between the distance of the QNG boards to the healer and the effect magnitude ($r = -0.18$, p = 0.30), indicating that the effect was spatially uniform rather than graded by proximity to the healer or the client. Of greatest interest, the healing-specific phase compared between the healing and control sessions was tested at six temporal resolutions, spanning 50–200 samples (10–40 s at 5 Hz). The effect was significant at five of the six temporal resolution sizes (p ranging from 0.009 to 0.038). Fig. 5 visualizes this outcome at a 20 s window. Cohen's d for the healing phase was $d = -1.22$.

In physical terms, this means the quantum noise output became slightly but reliably less fractally complex during healing. That is, the voltage fluctuations developed subtle temporal correlations across time scales, as if adjacent samples were no longer fully independent. The signal became “smoother” and more predictable than expected for a purely random process.

Discussion

The combined results of six biofield healing sessions, compared to six matched sessions that did not involve healing, indicated that the random noise outputs of 1000 QNGs deviated significantly beyond chance during the healing phase. Corroborating negentropic deviations were observed in the same data using Higuchi fractal dimension metric. While only six sessions advocates for caution in interpreting these results, the large Cohen’s d in the healing phase ($d = -1.22$) suggests that this may be a robust effect.

If this were the only study indicating a negentropic effect during healing intention, these results might be conservatively dismissed as a fluke. However, similar results have been previously reported during healing trials with random data generated by truly random number generators (RNG) and with QNGs. One such study involved 110 different volunteers, each receiving a half-hour biofield session by one of 17 practitioners using different modalities.[14] A second study consisted of 8 healing sessions contributed by four practitioners using the same modality (Johrei) over a three day period.[12] A third used an RNG to see if it would respond to a “healing presence”[20] Conceptually similar studies examining negentropic responses to small and large groups engaged in coherent mental activities, such as healing, meditation, and focused intention, have reported similar effects.[30–36]

Interpretation

How may we interpret these results? Each QNG sample measures the voltage output of a reverse-biased Zener diode. An increase in that voltage can indicate higher amounts of noise, possibly due to greater efficiency in the quantum tunnelling process. A shift from lower to higher noise amplitude, which is what this study observed during healing, also suggests the possibility of a modulation of ambient energy, potentially of the kind physicists and electrical engineers would recognize as conventional energy. Perhaps energy medicine practices involve some form of modulation of energy that is extracted from the environment and then redirected toward the client. Such energies may be associated with bioelectromagnetic effects[37], or with “free energies”.[38] This energetic interpretation is appealing because such a model might eventually be accommodated by existing mechanistic models of healing.

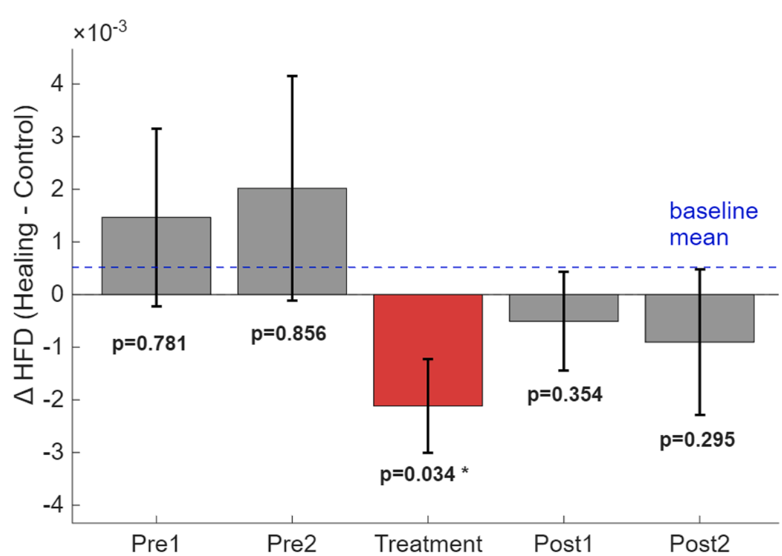

Fig. 5. This shows the difference in HFD in each phase of a session, compared between the healing and control data. The significant drop during the healing phase indicates that the QNG outputs became significantly “smoother,” or negentropic, in the healing sessions than in the control sessions.

This conjecture is attractive for the present results because it would help to explain why noise levels significantly *decreased* at the start of the healing period. That is, if no external energetics were imposed on the system, only ambient energies would remain available to manipulate. From a thermodynamic perspective, one could then model a healing encounter as a closed system. That is, to maintain an energetic equilibrium and increase energy in one location (or time), it would be necessary to decrease it elsewhere (or "else-when"). This idea, which would require something like a temporally flexible form of negentropy, is obviously speculative, but it is also testable.

Limitations

As a small-scale exploratory study, it is premature to generalize the present findings. In addition, the use of custom-designed QNG devices as a means of detecting entropic variations in the environment is a relatively new development, and unlike most commercially available RNGs, the outputs of such devices are not conditioned to strictly avoid environmental influences. I.e., electronic components can be temperature sensitive, so in future studies it would be important to simultaneously measure ambient temperature and adjust the outputs of the QNGs accordingly. Likewise, electromagnetic or magnetic fields may also influence QNGs, and in the present case these devices were located in a laboratory that included many other sensors and electronic equipment. It is possible, for example, that just the act of the healer entering and existing the laboratory during the pre- and post-healing period was sufficient to cause the significant swing in the QNG outputs, as seen in Fig. 4. Understanding if, and how, QNGs may interact with other electronic systems, and also the presence of human bodies, should be added to any future study design.

Ethics statement

This study was approved by Netherlands Organisation for Applied Scientific Research Institutional Review Board. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Generative AI statement

During the preparation of this work the authors used Claude Code v2.1.70, Opus 4.6, to assist with developing the MATLAB programs used in the data analyses. After using this tool, the authors reviewed the code to ensure its accuracy and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Dean Radin: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **John A. Ives:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

Dean Radin was an editorial board member of *Explore* at the time of submission, but this had no impact on the peer review process and the final decision. John A. Ives declares that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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